

Tianyue (Esther) Qian

tq23@duke.edu · github.com/Esther016 · linkedin.com/in/tianyue-qian/

EDUCATION

Duke University & Duke Kunshan University (DKU)

Expected Graduation: May 2028

Dual Degree Candidate (Major: Data Science)

- Current GPA: 3.87/4.00
- Honors: Dean's List with Distinction (Spring 2025), President's Scholarship (2023)
- Relevant Coursework: Natural Language Processing, Machine Learning, Probability & Statistics, Advanced Linear Algebra, Programming & Data Structures, Computer Organization

RESEARCH & PROJECTS

Fine-Grained Vision-Language Model Evaluation

Mar 2026 - Present

Independent Researcher | Advised by Prof. Kaizhu Huang

Kunshan, China

- Designed controlled evaluation experiments for Fine-R1 and its Qwen base model to examine robustness, consistency, and sensitivity to changes in evaluation conditions.
- Built Python-based evaluation pipelines for large-scale closed- and open-world testing and analyzed recurring failure patterns across experimental conditions.
- Compared model behavior under controlled perturbations to investigate whether observed performance changes reflected reasoning limitations, systematic response biases, or artifacts of the evaluation setup.

LLM Cultural Alignment - Computational Social Science Research

Sept 2025 – Mar 2026

Research Assistant (Independent Study) | [GitHub](#)

Kunshan, China

- Developed a model-agnostic bilingual LLM pipeline for large-scale prompting and structured response collection, incorporating multi-model API orchestration, retry and checkpointing mechanisms, statistical analysis, and visualization.
- Analyzed model behavior across languages, model origins, and prompting conditions to identify consistency, framing sensitivity, and cross-lingual differences.

TECHNICAL EXPERIENCE

Topcast Aviation Supplies Co., Ltd

Jun 2026 – Aug 2026

Business Transformation Intern

Hong Kong SAR

- Engineered an end-to-end Python/Selenium automation workflow for a recurring B2B aviation process, integrating Outlook-based MFA, configurable RFQ extraction, deduplication, SQL Server storage, and automated reporting.
- Reverse-engineered a complex ERP receipt workflow spanning 129 fields and multiple database tables; developed SQL Server simulation and validation scripts to reconstruct system logic, identify inconsistencies, and test process behavior.

Nanjing DeepGet Intelligent Technology Co., Ltd

Dec 2025 – Jan 2026

Technology Consulting Intern (Industrial AI / Tech Due Diligence)

Nanjing, China

- Synthesized technical meetings, internal research, and 20+ technical documents into structured memos assessing system capabilities, technical assumptions, implementation risks, and validation requirements for stakeholder decision-making.
- Evaluated an industrial computer-vision workflow across 500+ reviewed images, identified recurring failure patterns, and translated technical findings into practical observations on solution performance and deployment limitations.

TEACHING EXPERIENCE

Academic Resource Center (Duke Kunshan University)

Aug 2025 – May 2026

Peer Tutor for CS 201

Kunshan, China

- Facilitated weekly programming sessions for 10+ students, diagnosing Java and data-structure issues and communicating technical concepts through structured, problem-oriented explanations.

SKILLS

Programming & Data: Python, Java, SQL, pandas, NumPy, scikit-learn

Machine Learning & Analysis: Model Evaluation, Statistical Analysis, Matplotlib

Development & Systems: SQL Server, Selenium, Flask, REST APIs, Git/GitHub

Analytics: Power BI, Power Query, DAX

Languages: Mandarin Chinese (native), English (fluent)